

1. PROJECT OVERVIEW

- **Project Title:** FitBuddy - AI Workout Generator

2. FEATURE SUMMARY

The FitBuddy platform integrates several robust functional features designed to completely automate the fitness planning lifecycle while maintaining strict administrative oversight.

3. DETAILED FEATURE BREAKDOWN

- **Feature 1: Comprehensive User Input Interface**
 - *Description:* A responsive Jinja2/HTML form designed to capture detailed physical metrics, including the user's Name, Age, Weight, specific Fitness Goal (e.g., Weight Loss), and preferred Workout Intensity (e.g., Medium).
- **Feature 2: Dynamic 7-Day Plan Generation**
 - *Description:* Direct integration with the Google Gemini (gemini-3.5-flash-lite) Generative AI model to instantly synthesize a highly structured, day-by-day customized workout schedule alongside contextual nutrition tips.
- **Feature 3: Relational Data Persistence**
 - *Description:* Integration with a local SQLite database utilizing SQLAlchemy to persistently record newly registered user profiles and permanently store the original AI-generated fitness plans for future access.
- **Feature 4: Resilient Fallback Mechanisms**
 - *Description:* Robust backend exception handling featuring a strict 60-second timeout parameter (`request_options={"timeout": 60}`) to intercept network latency errors, ensuring the server does not crash during external AI outages.
- **Feature 5: Dynamic Web Rendering**

- *Description:* Utilization of Jinja2 templating to seamlessly inject the raw AI-generated text data into an elegant, readable HTML layout on the final result page.
- **Feature 6: Administrative Dashboard**
 - *Description:* A dedicated administrative view (/all-users) presenting a comprehensive tabular interface that displays all registered users (Name, Age, Weight, Goal, Intensity) alongside their corresponding generated workout plans.